

Task 20: The **Git** Journey

AI-Assisted Development & Professional Version Control

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1st DAW | Development Environments

Phase 1: **Programming** with AI

Human-AI collaboration to create robust application logic.

Logic Development

I utilized generative AI tools to structure a **Task Manager** in Java:

- Dynamic structure using ArrayList for storage.
- Data entry management via Scanner.
- **Robust Validation:** Implementation of try-catch to prevent data type errors.
- Interactive menu with switch-case logic.



Phase 2: Mastering Git CLI

Managing software lifecycle through the command line.

1. Project Initialization

The first step was transforming a local folder into a controlled repository:

```
$ git init  
Initialized empty Git repository in...
```

- Creation of the hidden `.git` directory.
- Preparation for version tracking.
- Local environment setup in IntelliJ IDEA.



2. Tracking and Staging



Current Status

Constant use of `git status` to identify modified or untracked files.



Preparation

Using `git add .` to move changes from the working directory to the Staging area.



Commit

Permanent record of changes using `git commit` with descriptive messages.

Control Points

07

Key Commands Mastered

During the workflow, essential commands were mastered to maintain a clean and coherent history, allowing rollbacks if necessary.

- `git log` to visualize the commit history.
- `git diff` to analyze specific line changes.
- `git branch -M main` to standardize branch naming.

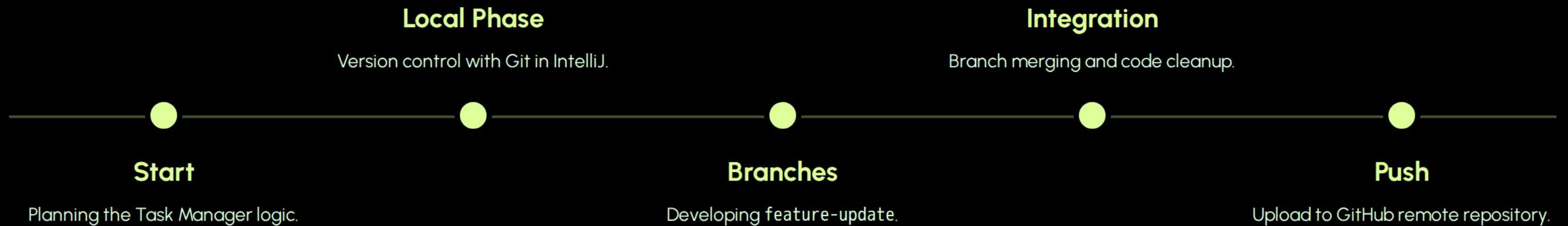


Branch Management

To implement new functions safely, a branching strategy was used:

- **Branching:** Creating the feature-update branch.
- **Isolation:** Developing improvements without affecting the stable main code.
- **Merge:** Successful fusion of the test branch into the principal branch.

The Project Roadmap



Deployment and Results

Taking the local project to the public cloud.

GitHub & Pages

Connecting the local repo with GitHub for cloud storage and activating **GitHub Pages** to host technical documentation via `index.html`.

Public Repository

Live Website

Any Questions?

Thank you for your attention.

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Image Sources



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